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1) What is Javascript? Why do we use Javascript in web development and what are its main features?

Javascript is a programming language used to make web pages interactive and dynamic.

We use Javascript to add functionality to websites, handle user actions, validate forms, and update content without reloading the page.

Main features of JS:

⇒ Lightweight: It is easy to learn and use.

⇒ Dynamic: Variables can store different types of values.

⇒ Interpreted: Js engines execute & optimize Js code

⇒ Event driven: It responds to user actions like clicks & typing

2) What is the difference b/w var, let, const in JS?

Features	Var	Let	Const
Scope	Function scope	Block scope	Block scope
Redeclaration	Allowed	Not allowed in same scope	Not allowed in same scope
Reassignment	Allowed	Allowed	Not allowed.
Initialization	Optional	optional	Required

3. What are the different data types in JS?

Define the types of value that a variable can store. JS has two main categories of data types Primitive & Non Primitive

Primitive Data types	Non Primitive Data types
String ⇒ Stores text	Object
Number ⇒ Stores numbers	Array
Boolean ⇒ Stores true or false	Function.
undefined ⇒ A variable declared without a value.	Non primitive data types are used to store collection or more complex data.
Null ⇒ Represents an intentional empty value.	
BigInt ⇒ Stores very large integer.	
Symbols ⇒ Represents a unique identifier.	

4. What is a function in JS? Explain the difference blw normal function, function expressions & arrow function

A function is a block of reusable code that Performs a particular task. It helps reduce code repetition.

There are three common ways to create functions.

⇒ Normal function.

⇒ Function Expression.

⇒ Arrow Function.

1. Normal Function:

A normal function is declared using the function keyword.

Ex:

```
function add(){
    console.log(10+20)
}
add();
```

2. Function Expression:

A function is stored inside a variable

Ex:

```
const add = function(){
    console.log(10+20);
}
add();
```

3. Arrow Function:

An arrow function is a shorter way of writing a function using the => symbol

Ex:

```
const add = () =>{
    console.log(10+20)
}
add();
```

5] What is the difference b/w == and === in JS? Explain with Examples?

Both == and === are comparison operator used to Compare Values.

Double equal (==):

It Compares Values after performing type conversation when necessary.

Ex:

```
console.log (100 == "100")
```

Triple Equal (===):

It Compares both value and data type without converting the types.

Ex:

```
console.log (10 === "10")
```

6] What is ReactJs? why do we use React instead of plain JS, and what are the main features of React?

ReactJs is a JS library used to build interactive and dynamic user interfaces, especially for web applications.

Why do we use React?

It makes UI development easier

It supports reusable components.

It updates the UI efficiently.

It helps manage application data using State.

It makes large applications easier to maintain

Main features of React:

1. Components => Divide the application into small Reusable parts.
2. JSX => Allows us to write HTML like syntax inside JS.
3. Virtual Dom => Helps update the UI efficiently.
4. Props => Used to pass data b/w Components.
5. State => Used to manage changing data
6. One way data flow => Data generally follows from Parent components to child components through props.
7. Hooks => Allows functional Components to use React features like state & effects.

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1. What is a Component in React? Explain functional Component & why React application use reusable components.

A component is a small, independent reusable Part of the a React application.

Components help divide a large application into smaller parts. A

Functional Component:

A functional Component is a JS function that returns JSX to display UI.

Why do we use reusable Components?:

Code reusability	Reduces code repetition.
Easy to maintain	Easy to understand

Q1 What is JSX in React? Why do we use JSX, and how is JSX different from HTML?

JSX (Javascript XML) is a syntax extension for JS that allows us to write HTML like code inside JS.

It commonly used in React to describe the user interface.

Why do we use JSX?

=> Easy to write UI code => Makes code readable

=> Combines JS and UI description => supports JS expression inside {}

HTML	JSX
used to create webpage structure.	used to describe React UI.
uses class	uses className
uses for	uses htmlFor
Event names are lowercase, like onClick	uses {} for Event names like onClick
can have multiple root elements in a document	A component must return one root element.

Q) What is the difference b/w Props & State in React? Explain how data is passed b/w Components & how State updates the UI.

Props and State are used to manage data in React applications.

Props:

Props are used to pass data from a parent Component to a child Component.

Ex:

Parent:

```
const App = () => {
  return (<>
  <student name = "Ram">
}
export default App
```

Child:

```
const student = (props) => {
  return (<>
    <h1> Hello {props.name} </h1>
}
export default Student
```

State:

State is used to store and manage data inside a Component. When state changes, React updates the UI.

Props.	State.
Used to pass data b/w components.	used to manage component data.
Passed from parent to child	Managed inside a Component
Read only for the receiving Component	updated using a state setter.
Helps share data	Helps update UI dynamically

10) What is the Virtual DOM in React? Explain the difference b/w Real DOM & Virtual DOM & how React updates the UI.?

Virtual DOM:

Is a lightweight JS representation of the Real DOM. React uses it to identify changes in the UI & update the necessary parts of the actual DOM.

Real DOM:

Real DOM represents the actual webpage elements displayed in the browser.

How React updates the UI:

=> Initially, React creates the UI.

=> React maintains a virtual DOM representation

=> When state or props change, React renders a new UI representation.

=> React compares the new representation with the previous one.

=> React commits the necessary changes to the Real DOM.